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标题	Microwave-Assisted Buchwald – Hartwig Double Amination: A Rapid and Promising Approach for the Synthesis of TADF Compounds
作者	Mohd Jamel N., Skhirtladze L., Hussein A., Ma Y., Woon K., Abdulwahab M. 等 8 人
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摘要（原文节选）

We herein report a microwave-assisted Buchwald – Hartwig double amination reaction to synthesize potential thermally activated delayed fluorescence compounds, forming C(sp2)-N bonds between donor and acceptor units. Our approach reduces reaction times from 24 h to 10 – 30 min and achieves moderate to excellent yields, outperforming conventional heating methods. The method is compatible with various aryl bromides and secondary amines, including phenoxazine, phenothiazine, acridine, and carbazole. Density functional theo ⋯

数据表 · Table 1（8 行，含数值行 8 行）

Optimization of the Buchwald – Hartwig Double Amination Reaction of Phenoxazine (PO-H) with 1,4-Dibromobenzene, 9(a), Assisted by Microwave Irradiation

entry	temperature (T/oC)	time (t/min)	products	yield (%)
1	130	30	14(a)	91
2	140	30	14(a)	90
3	150	30	14(a)	93
4	150	5	14(a)	80
5	150	10	14(a)	91
6	150	20	14(a)	90
7	150	25	14(a)	89
8	150	30	14(a)	91

正文对应句（逐字）：Our interest in the development of D – A – D TADF molecules has prompted us to explore methods that can be utilized for the formation of C(sp2)-N bonds between the donor and the acceptor. Carbazole (Cz-H) and its derivatives,3,26 − 29 10H-phenoxazine (PO-H)30 − 32 and 10H-phenothiazine (PS-H),31,33,34 and d

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